

Deferred/Supplementary Examination

Date : January 2023

Time : 2 hours

Marks available : 67

Full Marks : $66\frac{2}{3}$

You may answer the questions in any order, provided your solutions are clearly labelled.

Full answers are expected. Please write neatly.

The use of electronic devices is forbidden; this includes calculators.

1. For each of the following, either prove that the statement is true or give a reason why it is false.

(a) The terms of the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ can be rearranged to give any sum that you please.

(b) Every countable subset of $\mathbb{R}$ has a limit point.

(c) Every uncountable subset of $\mathbb{R}$ has a limit point.

(d) If a power series $\sum_{k=0}^{\infty} a_k x^k$ converges at some point $x_1 \neq 0$, then it also converges at $-x_1$.

[10]

2. Let $(a_k)_{k=1}^{\infty}$ be the sequence defined by
$$\begin{cases} a_1 = 1, \\ a_{k+1} = 3 + \left(\frac{a_k}{4}\right)^2 \end{cases} \quad \text{for all } k \in \mathbb{N}.$$

(a) Prove that $1 \leq a_k \leq a_{k+1} \leq 6$ for all $k \in \mathbb{N}$.

(b) Which theorem allows you to deduce that $(a_k)_{k=0}^{\infty}$ converges? Give the statement of this theorem and explain why it applies in this case.

(c) Find $\lim_{k \rightarrow \infty} a_k$.

[5, 4, 4]

3. Which of the following series are convergent? Justify your answers.

(a) $\sum_{k=0}^{\infty} \frac{4^k(k+1)}{k!}$

(b) $\sum_{k=1}^{\infty} \frac{\arctan(k)}{\sqrt{k^2+1}}$

[8]

4. For each of the following, determine whether the given series is absolutely convergent, conditionally convergent or divergent:

(a) $\sum_{k=1}^{\infty} \sin^k\left(k\frac{\pi}{6}\right)$

(b) $\sum_{k=0}^{\infty} (-1)^k \sin\left(\left(\frac{1}{2}\right)^k\right)$

(c) $\sum_{k=0}^{\infty} \frac{(-1)^k}{\sqrt[3]{k^2+8}}$

In each case, give full reasons for your answer.

[14]

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with $\lim_{x \rightarrow 2} f(x) = 3$. Prove *directly from the definition* that

$$\lim_{x \rightarrow 2} \frac{f(x)}{1 + f(x)} = \frac{3}{4}.$$

[6]

6. Let $f(x)$ be a continuous function on $\mathbb{R}$.

(a) Give an example to show that even if the set $A \subseteq \mathbb{R}$ is open, the set $\{f(x) : x \in A\}$ need not be open.

(b) Show that for any open set $B \subseteq \mathbb{R}$, the set $\{x \in \mathbb{R} : f(x) \in B\}$ is open. [2, 7]

7.(a) Prove that the series $\sum_{k=0}^{\infty} \frac{x^k}{1+x^k}$ converges absolutely and uniformly on the interval $[-r, r]$ for any real number r satisfying $0 < r < 1$.

(b) Explain why this series doesn't converge when $x = -1$.

(c) Why doesn't this series converge when $x = 1$? [5, 1, 1]
